

IBM HR Analytics– Project Requirements

TOOLS USED

- Excel (Pivot Tables, Dashboards)
- SQL (Data Querying & Analysis)
- Machine Learning (Python - Random Forest, Logistic Regression)

PROJECT OBJECTIVE

Create an all-inclusive HR Analytics project for analyzing employee attrition and performance data from IBM. The project should concentrate on attrition patterns, key factors influencing employee turnover, department-wise performance, and future attrition predictions.

PROBLEM STATEMENT

The project aims to answer the following business questions:

1. Attrition Overview

- What is the overall employee attrition rate?
- How many employees left vs stayed?
- What is the month-on-month attrition trend?

2. Demographic Analysis

- Which age group has the highest attrition?
- Is there a gender difference in attrition?
- Which education level shows highest turnover?
- How does marital status affect attrition?

3. Department & Role Analysis

- Which department has highest attrition?
- Which job role has highest turnover?
- How does job level affect attrition?

4. Job Satisfaction Analysis

- How does job satisfaction impact attrition?
- What is the relationship between environment satisfaction and attrition?
- How does job involvement affect retention?

5. Work-Life Balance Analysis

- How does work-life balance affect attrition?
- What is the impact of overtime on employee turnover?
- How does distance from home affect attrition?

6. Compensation Analysis

- How does monthly income affect attrition?
- What is the income difference between leavers and stayers?
- How does percent salary hike impact retention?
- How does stock option level affect attrition?

7. Tenure Analysis

- How do years at company affect attrition?
- What is the impact of years since last promotion?
- How does number of companies worked affect attrition?

8. Predictive Analytics (ML)

- Can we predict which employees are likely to leave?
- What are the top predictors of attrition?
- How accurate can we predict attrition?

1. TOTAL ATTRITION ANALYSIS

Requirements:

- Calculate overall attrition rate (Yes/No percentage)
- Determine total employees (1,470)
- Calculate number of employees who left (237)
- Calculate number of employees who stayed (1,233)
- Compare attrition rate by department

Sample KPIs:

- Total Employees: 1,470
- Attrition Rate: 16.12%
- Employees Left: 237
- Employees Stayed: 1,233

2. DEMOGRAPHIC ATTRITION ANALYSIS

Requirements:

- Calculate attrition by Age Group (18-25, 26-35, 36-45, 46-55, 56+)
- Calculate attrition by Gender (Male vs Female)
- Calculate attrition by Education Level (Below College, College, Bachelor, Master, Doctor)
- Calculate attrition by Marital Status (Single, Married, Divorced)

Sample Output:

- Age Group 18-25: 35.77% attrition (HIGHEST)
- Age Group 36-45: 9.19% attrition (LOWEST)
- Male Attrition: 17%
- Female Attrition: 15%
- Bachelor Degree: 20% attrition (HIGHEST)
- Doctorate: 10% attrition (LOWEST)

3. DEPARTMENT & JOB ROLE ANALYSIS

Requirements:

- Calculate attrition by Department (Sales, R&D, HR)
- Calculate attrition by Job Role (Sales Executive, Research Scientist, Lab Technician, etc.)
- Identify top 3 highest attrition job roles
- Calculate attrition by Job Level (1-5)

Sample Output:

- Sales Department: 20.63% attrition (HIGHEST)
- R&D Department: 13.84% attrition (LOWEST)
- Sales Representative: 40% attrition (HIGHEST)
- Job Level 1: Highest attrition
- Job Level 5: Lowest attrition
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4. JOB SATISFACTION ANALYSIS

Requirements:

- Calculate attrition by Job Satisfaction (Low, Medium, High, Very High)
- Calculate attrition by Environment Satisfaction (Low, Medium, High, Very High)
- Calculate attrition by Relationship Satisfaction
- Calculate attrition by Job Involvement

Sample Output:

- Low Job Satisfaction: 22.84% attrition
- Very High Job Satisfaction: 11.33% attrition
- Low Environment Satisfaction: 25% attrition

5. WORK-LIFE BALANCE & OVERTIME ANALYSIS

Requirements:

- Calculate attrition by Work-Life Balance (Bad, Good, Better, Best)
- Calculate attrition by Overtime (Yes vs No)
- Calculate attrition by Distance From Home (buckets)
- Calculate attrition by Business Travel frequency

Sample Output:

- Bad Work-Life Balance: 31.25% attrition
- With Overtime: 30.53% attrition
- Without Overtime: 10.44% attrition (3x lower!)
- Frequent Travel: 25% attrition

6. COMPENSATION ANALYSIS

Requirements:

- Calculate average monthly income of stayers vs leavers
- Calculate income by education level and attrition
- Calculate attrition by Percent Salary Hike
- Calculate attrition by Stock Option Level (0,1,2,3)

Sample Output:

- Stayers Avg Income: \$6,833
- Leavers Avg Income: \$4,787 (Difference: \$2,046)
- No Stock Option: 24.41% attrition
- Stock Option Level 2: 7.59% attrition

7. TENURE & PROMOTION ANALYSIS

Requirements:

- Calculate attrition by Years at Company
- Calculate attrition by Years Since Last Promotion
- Calculate attrition by Years in Current Role
- Calculate attrition by Years With Current Manager
- Calculate attrition by Number of Companies Worked

Sample Output:

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- 1-2 Years at Company: Highest attrition
- No Promotion: 24% attrition
- 5+ Years at Company: Lower attrition

8. DASHBOARD REQUIREMENTS (EXCEL)

Create an interactive dashboard containing:

KPI Cards:

- Total Employees (1,470)
- Attrition Rate (16.12%)
- Total Left (237)
- Total Stayed (1,233)
- Avg Income Leavers (\$4,787)
- Avg Age Leavers (33.6)

9. SQL REQUIREMENTS

Use SQL queries to generate insights (31 queries provided):

Basic Analysis:

- Total employees, leavers, stayers
- Attrition rate calculation
- Attrition by department, age, gender

Advanced Analysis:

- Attrition by education level
- Job satisfaction impact
- Work-life balance impact
- Overtime impact analysis

Aggregations:

- Income comparison (AVG, MIN, MAX)
- Years at company analysis
- Years since last promotion analysis
- Stock option level analysis

Risk Identification:

- High-risk employees (overlap of multiple factors)
- Department-wise risk factors
- At-risk employees for HR intervention

10. MACHINE LEARNING REQUIREMENTS

Build a classification model to predict employee attrition

Task Type: Binary Classification (Will employee leave? Yes/No)

Target Variable: Attrition (Yes/No)

Features Used (35 columns):

- Age, Gender, Department, JobRole
- MonthlyIncome, PercentSalaryHike, StockOptionLevel
- YearsAtCompany, YearsSinceLastPromotion
- JobSatisfaction, WorkLifeBalance, OverTime
- DistanceFromHome, Education, MaritalStatus

Suggested Models:

- Logistic Regression (Baseline)
- Decision Tree Classifier
- Random Forest Classifier (Best performing)
- Gradient Boosting Classifier
- Support Vector Machine (SVM)

Model Evaluation Metrics:

- Accuracy Score
- Precision Score
- Recall Score
- F1 Score
- ROC-AUC Score
- Confusion Matrix

Expected Output:

- Best Model: Random Forest Classifier
- Accuracy: 85%
- ROC-AUC: 0.78
- Top Features: Overtime, Job Satisfaction, Years at Company, Monthly Income, Age

Prediction Capability:

- Identify employees at high risk of leaving
- Probability score for each employee
- Feature importance ranking